

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2014

CL306 Transportation Engineering-II

Date: 06.05.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Classify the Station Yards. [02]
(b) Discuss the requirements of an ideal Permanent Way. [05]
- Q - 2 (a) Explain scissor cross over with neat sketch [04]
(b) Discuss the necessity of Curves in Railways. Also write a short note on Types of curves. [05]
(c) What do you understand by geometric design of a railway track? Discuss in brief the parameters which affect it. [05]

OR

- Q - 2 (a) Draw neat sketch of diamond crossing in railways, indicating the important features. [04]
(b) Define Superelevation and mention the objectives of providing superelevation to Railway Tracks. Write the formulae of superelevation for B.G, M.G. & N.G. Also, discuss 'Negative Superelevation' with neat diagram. [05]
(c) What is cant? Explain cant excess and cant deficiency with their values for railway track. [05]
- Q - 3 (a) Enumerate advantages and disadvantages of railway transport as compared to other modes of transport. [02]
(b) Discuss about various fittings and fixtures used in railway track construction. [04]
(c) Do as Directed:
(i) Draw a neat sketch of semaphore signal. [03]
(ii) Enlist different types of light signals and explain any one. [05]

OR

- Q - 3 (a) Give a brief classification of Indian Railways. [02]
(b) Explain in brief plate laying and laying of ballast on track. [04]
(c) Do as Directed:
(i) List out different types of Signals used in Indian Railways. [03]

- (ii) Write a note on ware house and transit sheds with neat sketch. [05]

SECTION – II

- Q - 4 (a) What are drawbacks and benefits of water transportation? [04]
(b) How to construct rubble mound breakwater? [03]
- Q - 5 (a) Write a note on minimum turning radius with neat sketch. [04]
(b) According to Federal Aviation Administration, gives recommended phases of master plan. [05]
(c) Define maximum gross take-off weight and explain noise effect due to aircraft. [05]

OR

- Q - 5 (a) Discuss how air transportation better than the other modes of transportation. [04]
(b) According to Federal Aviation Administration, gives objectives of master plan. [05]
(c) Why we need to control air traffic? [05]
- Q - 6 (a) Define: stop way, cross wind. [02]
(b) The length of runway at sea level under standard condition at zero gradient is 1500 m. the airport site at an elevation of 900 m, the reference temperature is 20 ° C and the proposed runway permits ascending gradient of 0.5%. design the longitudinal profile of runway. [07]
(c) Enlist factors affecting runway orientation. [05]

OR

- Q - 6 (a) In case of basic runway length, explain normal landing case. [02]
(b) Find the design length of runway required if the length under standard atmospheric conditions is 2000 m, the actual elevation of the site is 1000 m above MSL and the airport reference temperature is 16° C. maximum effective gradient is 0.5%. [07]
(c) Explain wind rose diagram. [05]
